TERMWISE ISOMORPHIC QUOTIENTS GIVE ISOMORPHIC I-ADIC COMPLETIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The inverse system of isomorphism sets Isom_A(M_n, N_n) satisfies the Mittag-Leffler condition. invoked at proof lines 29132-29133. Makes the limit of nonempty sets nonempty.
- An isomorphism of the system of quotients M/I^n M and N/I^n N determines and is determined by an isomorphism of the completions. invoked at proof lines 29135-29136. Converts the nonempty limit into an isomorphism of completions.

Source reference. Original lemma 09BB: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29060–29060. Cited at proof lines 29132-29133 for the Mittag-Leffler property; the conclusion is used inside the counted span.

Lemma 0.2. *Let A be a Noetherian ring. Let $I \subset A$ be an ideal. Let M, N be finite A-modules. Set $M_n = M/I^nM$ and $N_n = N/I^nN$. If $M_n \cong N_n$ for all n, then $M^\wedge \cong N^\wedge$ as $A^\wedge$-modules.*

Proof. By Lemma 09BB the system $(\mathrm{Isom}_A(M_n, N_n))$ is Mittag-Leffler. By assumption each of the sets $\mathrm{Isom}_A(M_n, N_n)$ is nonempty. Hence $\lim \mathrm{Isom}_A(M_n, N_n)$ is nonempty. Since $\lim \mathrm{Isom}_A(M_n, N_n) = \mathrm{Isom}_{A^\wedge}(M^\wedge, N^\wedge)$ we obtain an isomorphism. □

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